



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 06 – “Type II”

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Previously, intermediate state

An inhomogeneous state arising from demagnetising field.

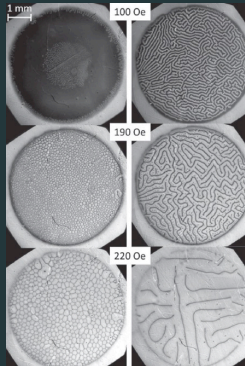


Figure 1: Magneto-optical images (using Faraday rotation) of an intermediate state of a Pb single crystal at 5 K in different magnetic fields [R. Prozorov, Phys. Rev. Lett., 98, 257001 (2007)].

Interface energy

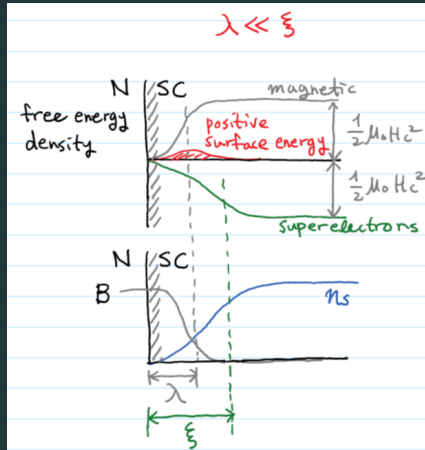


Figure 2: Magnetic flux and superconducting electron density profile across the normal metal-superconductor boundary

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Surface energy

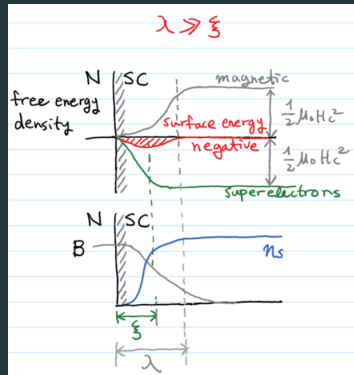
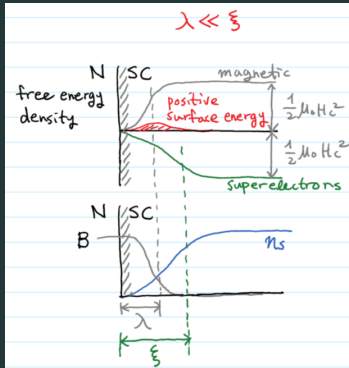
Magnetic field response

Vortices and flux lattice

Surface energy

Intermediate vs. Mixed

- Due to demagnetising field;
- Structure $\gg \lambda$ or ξ .

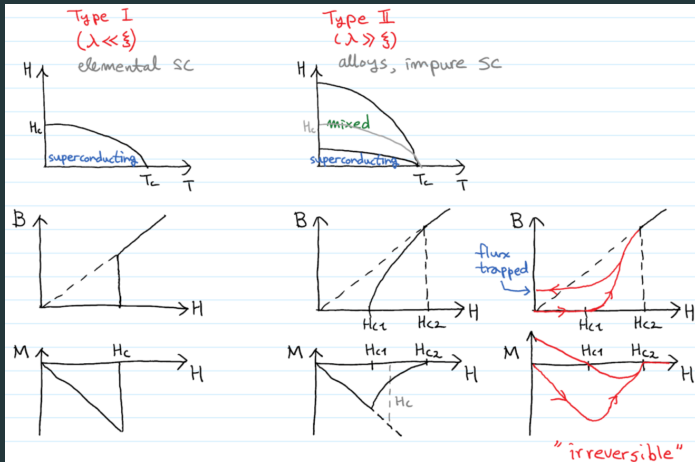


Let's assume an infinitely long rod in a longitudinal field unless stated otherwise.

Magnetic field response

Experimental magnetisation

Superconductors fall into two categories from their field response.



Mixed state from negative surface energy

- Type-II behaviour observed and attributed as dirty effects,
- Only later, a link to negative surface energy found,
- Let's see what's happening microscopically.

Vortices and flux lattice

Negative surface energy favours

- as many normal-superconducting interfaces as possible,
- as large surface area per volume as possible
→ rod-like structure with shrinking radius.

Vortices

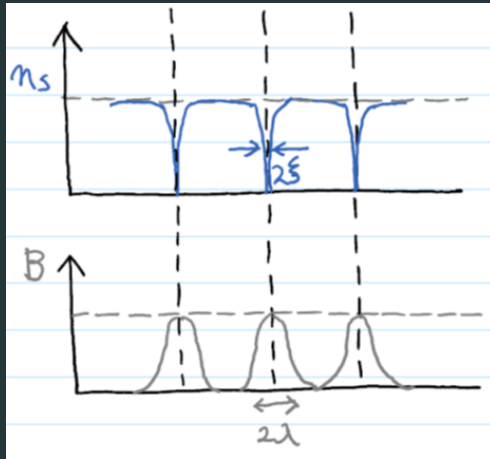


Figure 5: Schematic cross section of vortices.

Flux lattice

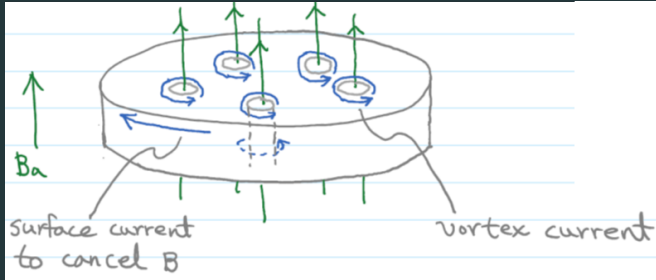
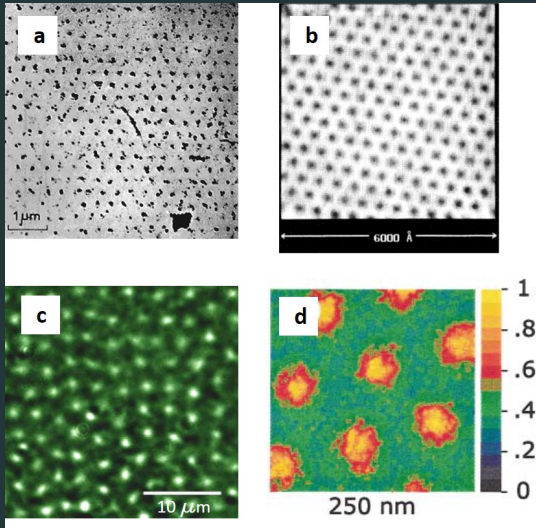


Figure 6: Schematic drawing of a flux lattice.

Experimental flux lattices



Experimental flux lattices (captioned)

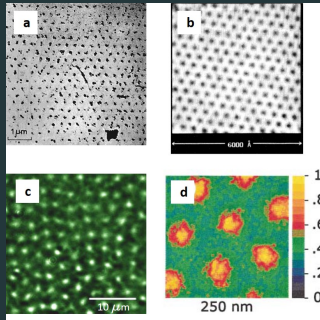


Figure 7: Flux lattices of mixed state in different materials. (a) Electron microscope image of Pb-In at 1.1 K in a field of 3 kOe [Essmann 1967](b) Scanning tunneling microscopic image of NbSe₂ [PhysRevLett.62.214]. (c)Magneto-optical image of NbSe₂ at 4 K in earth field [Goa 2001]. (d)Scanning tunneling spectroscopic image of MgB₂ at 2 K in a field of 0.05 T [PhysRevLett.89.187003].

Type-I vs Type-II

Type-I: almost all pure elemental superconductors.

- A reversible first-order transition with a latent heat across H_c and
- an intermediate state.

Type-II: alloys or impure metals.

- Contains trapped magnetic flux,
- Show a large magnetic hysteresis,
- Remains superconducting at fields much greater than the thermodynamic H_c one would expect from the heat capacity

Coming up next

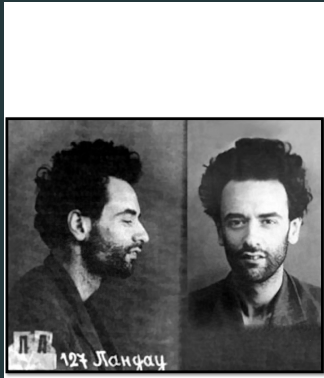


Figure 8: A mug shot of young Lev D. Landau in prison, 1938-9. What has happened? Landau developed masterful theories of metal, diamagnetism, superfluid, superconductor and phase transition, to name a few! (Photo from Wikipedia)